

FileEditSelectionViewGoRunTerminalHelpPython\_Test

EXPLORER

PYTHON\_TEST

create\_challenge\_images.py

create\_test\_image\_day3.py

create\_test\_image.py

crop.py

custom\_cyberpunk.png

custom\_mask.png

custom\_sketch.png

custom\_vintage.png

day5\_transformation.py

day6\_arithmetic\_bitwise.py

day7\_mini\_challenge.py

day8\_colour\_spaces.py

day9\_histogram.py

day10\_smoothing\_blurring.py

day11\_thresholding.py

day12\_gradients\_edges.py

dissolve.mp4

drawin\_basics.py

drawing.py

edge\_preserving\_comparison.png

equalization\_comparison.png

equalize.py

equalized\_image.png

ex\_1\_day\_3.py

ex\_2\_day\_3.py

ex\_3\_day\_3.py

ex\_4\_day\_3.py

exercise\_1\_custom\_pipeline.py

exercise\_1\_house.py

exercise\_1.py

exercise\_2\_comparison.py

OUTLINE

TIMELINE

ole\_thresholding.py

otsu\_and\_riddler.py

adaptive\_thresholding.py

day12\_gradients\_edges.py

exercise1\_in

CHAT

day12\_gradients\_edges.py

191 print("-" \* 50)

192 def interactive\_canny(image\_gray):

193 """Interactive tool to find optimal Canny parameters"""

194 # Apply initial blur (always recommended)

195 blurred = cv2.GaussianBlur(image\_gray, (5, 5), 0)

196

197 cv2.namedWindow("Canny Explorer")

198

199 # Create trackbars

200 cv2.createTrackbar("Low Threshold", "Canny Explorer", 50, 255, lambda x: None)

201 cv2.createTrackbar("High Threshold", "Canny Explorer", 150, 255, lambda x: None)

202 cv2.createTrackbar("Blur Size", "Canny Explorer", 5, 21, lambda x: None)

203

204 print("\nCanny Explorer Controls:")

205 print(" - Low Threshold: edges below this are ignored")

206 print(" - High Threshold: edges above this are definitely edges")

207 print(" - Blur Size: noise reduction (must be odd)")

208 print(" - Press ESC to exit")

209 print("\nRule of thumb: High = 2x to 3x Low")

210

211 while True:

212 low = cv2.getTrackbarPos("Low Threshold", "Canny Explorer")

213 high = cv2.getTrackbarPos("High Threshold", "Canny Explorer")

214 blur\_size = cv2.getTrackbarPos("Blur Size", "Canny Explorer")

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

--- Why blur before Canny? ---

CONCLUSION: ALWAYS BLUR BEFORE CANNY!

- Without blur: many false edges from noise

- With blur: clean, meaningful edges

---

--- Canny on Test Image ---

Canny produces clean, thin, continuous edges!

[Section 5] Interactive Canny Parameter Explorer

-----

Canny Explorer Controls:

+ v ... | {} x

powershell

powershell

powershell

powershell

powershell

powershell

powershell

powershell

powershell

powershell

+ day12\_gradients\_edges.py

Describe what to build

+ | Auto

↑

Local

Default Approvals

Ln 205, Col 60

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UTF-8

CRLF

{ } Python

16:39

17-06-2026

Type here to search

35°C

Partly sunny

ENG

15

EXPLORER

- PYTHON\_TEST
  - day10\_smoothing\_blurring.py
  - day11\_thresholding.py
  - day12\_gradients\_edges.py
  - dissolve.mp4
  - drawin\_basics.py
  - drawing.py
  - edge\_preserving\_comparison.png
  - equalization\_comparison.png
  - equalize.py
  - equalized\_image.png
  - ex\_1\_day\_3.py
  - ex\_2\_day\_3.py
  - ex\_3\_day\_3.py
  - ex\_4\_day\_3.py
  - exercise\_1\_custom\_pipeline.py
  - exercise\_1\_directional\_edges.py**
  - exercise\_1\_house.py
  - exercise\_1.py
  - exercise\_2\_comparison.py
  - exercise\_2\_target.py
  - exercise\_2.py
  - exercise\_3\_chessboard.py
  - exercise\_3.py
  - exercise\_4\_annotation.py
  - exercise\_5\_clock.py
  - exercise1\_brightness\_contrast.py
  - exercise1\_channel\_art.py
  - exercise1\_histogram\_analyzer.py
  - exercise1\_interactive\_blur.py
  - exercise1\_threshold\_optimizer.py
  - exercise1\_thumbnail\_gallery.py
- OUTLINE
- TIMELINE

```

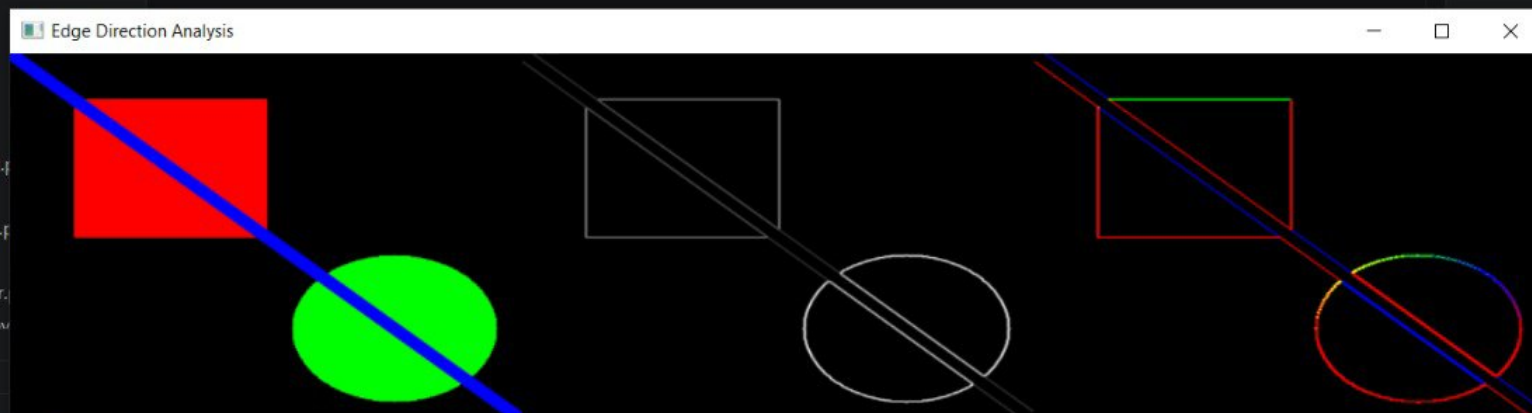
otsu_and_riddler.py adaptive_thresholding.py day12_gradients_edges.py exercise_1_directional_edges.py
exercise_1_directional_edges.py
54
55 if image is not None:
56     direction_vis, magnitude = analyze_edges(image)
57
58     # Convert magnitude to BGR for stacking
59     magnitude_bgr = cv2.cvtColor(magnitude, cv2.COLOR_GRAY2BGR)
60
61     comparison = np.hstack([
62         cv2.resize(image, (350, 250)),
63         cv2.resize(magnitude_bgr, (350, 250)),
64         cv2.resize(direction_vis, (350, 250))
65     ])
66
67     cv2.imshow("Edge Direction Analysis", comparison)
68     cv2.waitKey(0)
69
70     cv2.imwrite("edge_directions.png", direction_vis)
71
72 cv2.destroyAllWindows()

```

CHAT

Build with Agent

AI responses may be inaccurate  
Generate Agent Instructions to onboard AI onto your codebase.



File Edit Selection View Go Run Terminal Help Python\_Test

EXPLORER

- PYTHON\_TEST
  - drawin\_basics.py
  - drawing.py
  - edge\_directions.png
  - edge\_preserving\_comparison.png
  - equalization\_comparison.png
  - equalize.py
  - equalized\_image.png
  - ex\_1\_day\_3.py
  - ex\_2\_day\_3.py
  - ex\_3\_day\_3.py
  - ex\_4\_day\_3.py
  - exercise\_1\_custom\_pipeline.py

daptive\_thresholding.py day12\_gradients\_edges.py exercise\_1\_directional\_edges.py exercise\_2\_auto\_canny.py

```
exercise_2_auto_canny.py
69 comparison = np.hstack([
70     cv2.resize(gray_bgr, (250, 200)),
71     cv2.resize(manual_bgr, (250, 200)),
72     cv2.resize(auto_bgr, (250, 200))
73 ])
74
75 cv2.putText(comparison, "Original", (80, 30),
76             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)
77
78 cv2.putText(comparison, "Manual (50,150)", (300, 30),
79             cv2.FONT_HERSHEY_SIMPLEX, 0.5, (255, 255, 255), 1)
80
81 cv2.putText(comparison, "Auto Canny", (550, 30),
```

Auto Canny - sigma=0.2

sigma=0.20 (0,0)

Auto Canny - sigma=0.5

sigma=0.50 (0,0)

Auto Canny - sigma=0.33

sigma=0.33 (0,0)

ditional\_edges.py  
canny.py

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File Edit Selection View Go Run Terminal Help Python\_Test

EXPLORER

- PYTHON\_TEST
  - ex\_2\_day\_3.py
  - ex\_3\_day\_3.py
  - ex\_4\_day\_3.py
  - exercise\_1\_custom\_pipeline.py
  - exercise\_1\_directional\_edges.py
  - exercise\_1\_house.py
  - exercise\_1.py
  - exercise\_2\_auto\_canny.py
  - exercise\_2\_comparison.py
  - exercise\_2\_target.py
  - exercise\_2.py
  - exercise\_3\_chessboard.py
  - exercise\_3\_document\_edges.py
  - exercise\_3.py
  - exercise\_4\_annotation.py
  - exercise\_4\_sharpness\_analysis.py**
  - exercise\_5\_clock.py
  - exercise1\_brightness\_contrast.py
  - exercise1\_channel\_art.py
  - exercise1\_histogram\_analyzer.py
  - exercise1\_interactive\_blur.py
  - exercise1\_threshold\_optimizer.py
  - exercise1\_thumbnail\_gallery.py
  - exercise2\_color\_picker.py
  - exercise2\_document\_binimization.py
  - exercise2\_equalization\_grid.py
  - exercise2\_image\_dissolve.py
  - exercise2\_noise\_reduction.py
  - exercise2\_panorama\_prep.py
  - exercise3\_3d\_color\_histogram.py
  - exercise3\_color\_isolation.py

PROBLEMS OUTPUT

PS C:\Users\jebro... Automatic Canny w... Sigma=0.20: lower... Sigma=0.33: lower... Sigma=0.50: lower... Sigma=0.70: lower... PS C:\Users\jebro... Found document wi... PS C:\Users\jebro... PS C:\Users\jebro...

exercise\_4\_sharpness\_analysis.py

```
83 for i, (img, lap, name, var, status) in enumerate(results):
101     cv2.putText(
103         f"Score: {var:.0f}",
104         (10, y_start + 50),
105         cv2.FONT_HERSHEY_SIMPLEX,
106         0.4,
107         (200, 200, 200),
108         1
109     )
110
111     cv2.putText(
112         display,
113         f"Status: {status}",
114         (10, y_start + 70),
115         cv2.FONT_HERSHEY_SIMPLEX,
116         0.4,
117         (200, 200, 200),
118         1
119     )
120
121     cv2.imshow(name, img)
122     cv2.imwrite(f"{name}.png", img)
123
124     cv2.waitKey(1)
125     cv2.destroyAllWindows()
```

Image Sharpness Analysis

Original  
Score: 518  
Status: SHARP

Blurry  
Score: 1  
Status: BLURRY

Very Blurry  
Score: 1  
Status: BLURRY

Sharpness Score: 518  
Status: SHARP

Sharpness Score: 1  
Status: BLURRY

Sharpness Score: 1  
Status: BLURRY

Build with Agent

Be inaccurate  
tions to onboard AI  
odebase.

analysis.py

approvals

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17-06-2026

Python\_Test

File Edit Selection View Go Run Terminal Help

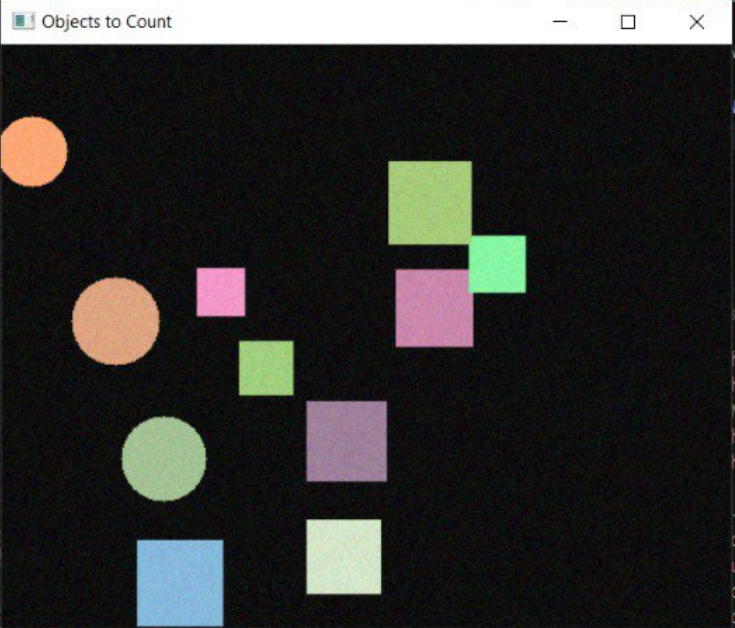
EXPLORER

- PYTHON\_TEST
  - ex\_4\_day\_3.py
  - exercise\_1\_custom\_pipeline.py
  - exercise\_1\_directional\_edges.py
  - exercise\_1\_house.py
  - exercise\_1.py
  - exercise\_2\_auto\_canny.py
  - exercise\_2\_comparison.py
  - exercise\_2\_target.py
  - exercise\_2.py
  - exercise\_3\_chessboard.py
  - exercise\_3\_document\_edges.py
  - exercise\_3.py

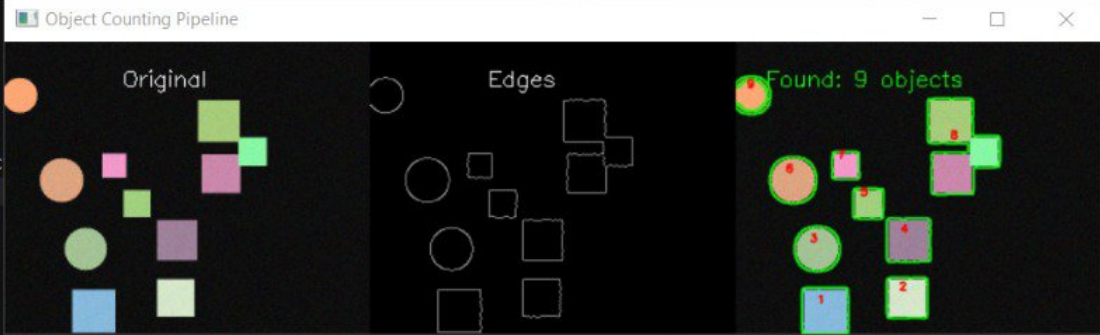
exercise\_5\_object\_counting.py

```
126 cv2.putText(  
128     f"Found: {count} objects",  
129     (520, 30),  
130     cv2.FONT_HERSHEY_SIMPLEX,  
131     0.5,  
132     (0, 255, 0),  
133     1  
134 )  
135  
136 cv2.imshow("Object Counting Pipeline", pipeline)  
137  
138 cv2.waitKey(0)  
139  
140 cv2.imwrite("object_counting.png", result)
```

Objects to Count



Object Counting Pipeline



DEBUG CONSOLE

TERMINAL

```
hell.PSConsoleReadLine.ForceRender()  
hell.PSConsoleReadLine.ProcessOneKey(ConsoleKeyInfo key, Dictionary`2 dispatchT  
NoAction, Object arg)  
hell.PSConsoleReadLine.InputLoop()  
hell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntri  
-----  
on\Desktop\Python_Test> python exercise_5_object_counting.py  
und 9 objects.  
on\Desktop\Python_Test> python^C  
on\Desktop\Python_Test> python exercise_5_object_counting.py
```

powerShell

python

Describe what to build

Local Default Approvals

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35°C Mostly sunny 16:53 17-06-2026